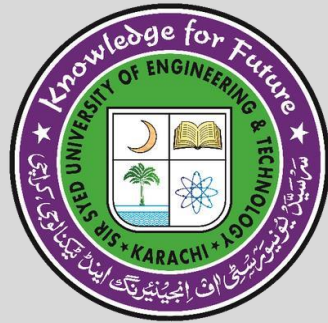




# **Final-Year Project Presentation**

## **Computer Engineering Department**



**SIR SYED UNIVERSITY OF ENGINEERING  
& TECHNOLOGY**



## Group Members

- **Zahir Khan (GL)** [2020-CE-121]
- **Adil Aijaz** [2020-CE-095]
- **Qasim** [2020-CE-113]
- **Ramish Masood** [2020-CE-086]



# AUGMENTED REALITY BASED COMPUTER FUNDAMENTALS COURSE

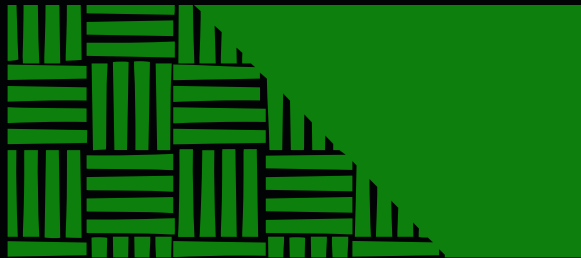
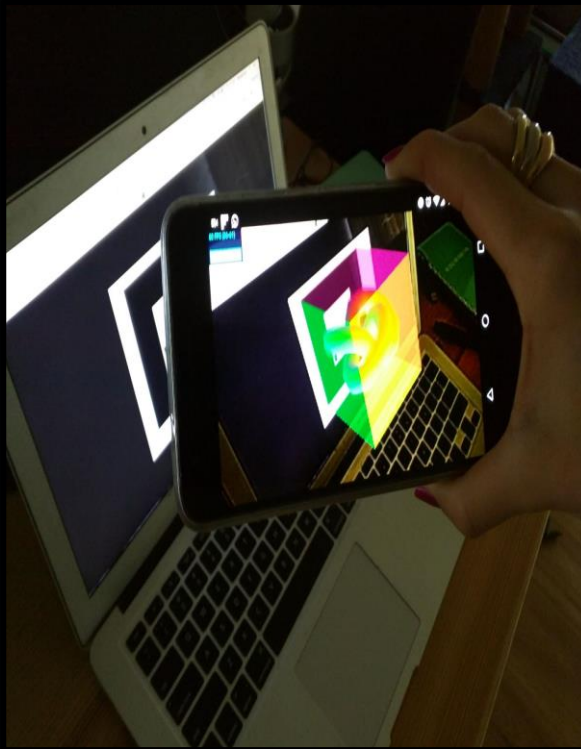
Our project aims to utilize AR technology to make visualizing computer components and devices more engaging and interactive for students. Traditional teaching methods like lectures and diagrams can be dry and challenging for students, but with our AR application, students can enjoy a more immersive learning experience by interacting with computer components in a virtual environment.



# Problem Identification



- **Boring Study Stuff:** Regular textbooks and lectures are dull and don't grab students' attention.
- **Hard to Understand Concepts:** Without pictures or interactive stuff, it's tough for students to understand tricky things.
- **Desire for Fun Learning:** Students want studying to be more fun and not just boring reading and listening.



## Need of Solution

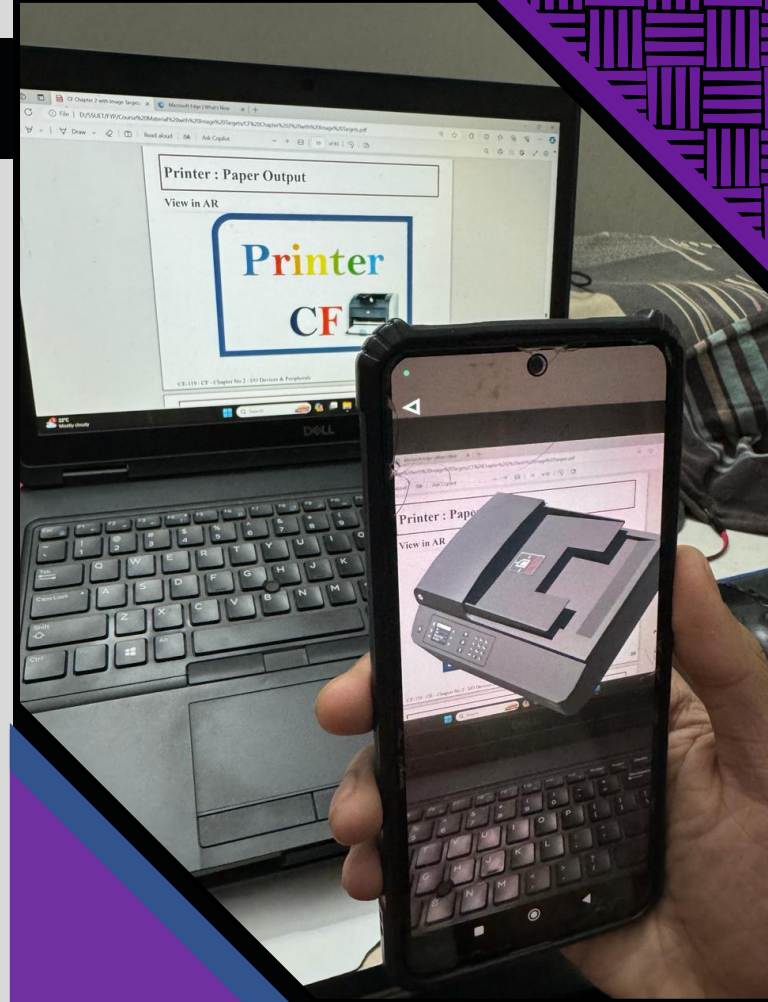


- **Student engagement and interest :** AR technology spices up the studying with interactive 3D models popping out of textbooks which increases their interest.
- **Visual Support:** Our AR solution provides visual aids to help students understand tough concepts.
- **Fun Learning Experience:** Studying is no longer dull with our AR technology. It's like bringing your textbooks to life, making learning enjoyable and unforgettable.



# Originality and Creativity

- Our aim is to assist students interested in learning about computer hardware and devices.
- We offer an interactive and engaging learning experience by utilizing 3D visualizations of computer components and devices through Augmented Reality (AR) technology.
- While AR has already found its place in the Medical and Automotive sectors, its adoption in the Education sector is still in its early stages, giving us a competitive edge over other providers.





# Project Value Proposition



## Interactive Learning Experience

Provides Interactive learning experience using 3D visualizations of computer components

## Accessibility & Availability

24/7 accessibility to learners of all ages

## Future-Readiness

Align students with future trends and advancements like AR technology

## Cost-Effectiveness

Reduces the need for physical hardware, making it a more cost-effective learning approach

## Enjoyable Learning

Enjoyable learning approach increase students' curiosity & motivation



# Software Tools & Programming Language



## **Blender**

For designing  
3D models and  
Editing 3D models  
imported from  
internet



## **Vuforia**

Provides SDK for  
AR and stores  
Image Targets of  
3D Models



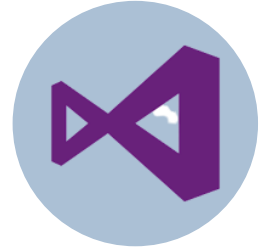
## **Unity**

Integrate image  
targets, C#  
scripts and apply  
textures &  
animation models



## **C#**

Used for writing  
scripts of  
rotation and  
zoom of 3D  
models



## **Visual Studio IDE**

Used to write C#  
script



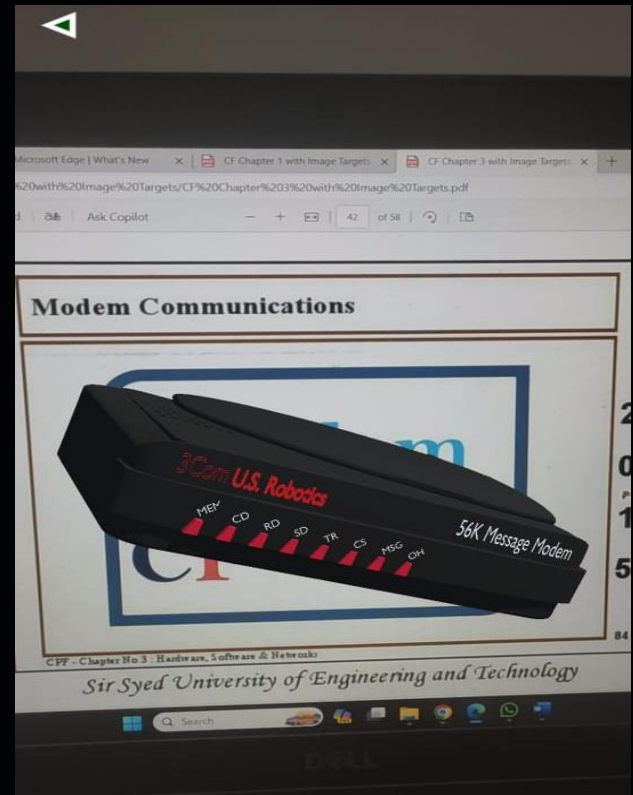
# Features

1

## View 3D model on AR Camera

Students can view 3D models in AR environment using phone camera

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# Features

2

## View 3D models manually

No image targets, no problem! Students can navigate between the 3D models using navigation





# Features

3

## Offline Access

No internet, No problem! Once the student download the App no internet is required to run the App

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# Features

4

## View & Download Lectures

Students can view and download lectures using AR-CF website

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12:00 PM | 26.0KB/s | 📶 | 37%

webcf.netlify.app/lectu

CF Chapter 2.pdf

Chapter No 2 : I/O Devices and Peripherals

**C E - 119**  
**Computing Fundamentals (CF)**

Compiled By:  
**Muhammad Ahsan Khan**  
[muzammil.saad@yahoo.com](mailto:muzammil.saad@yahoo.com)

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Computer Engineering Department  
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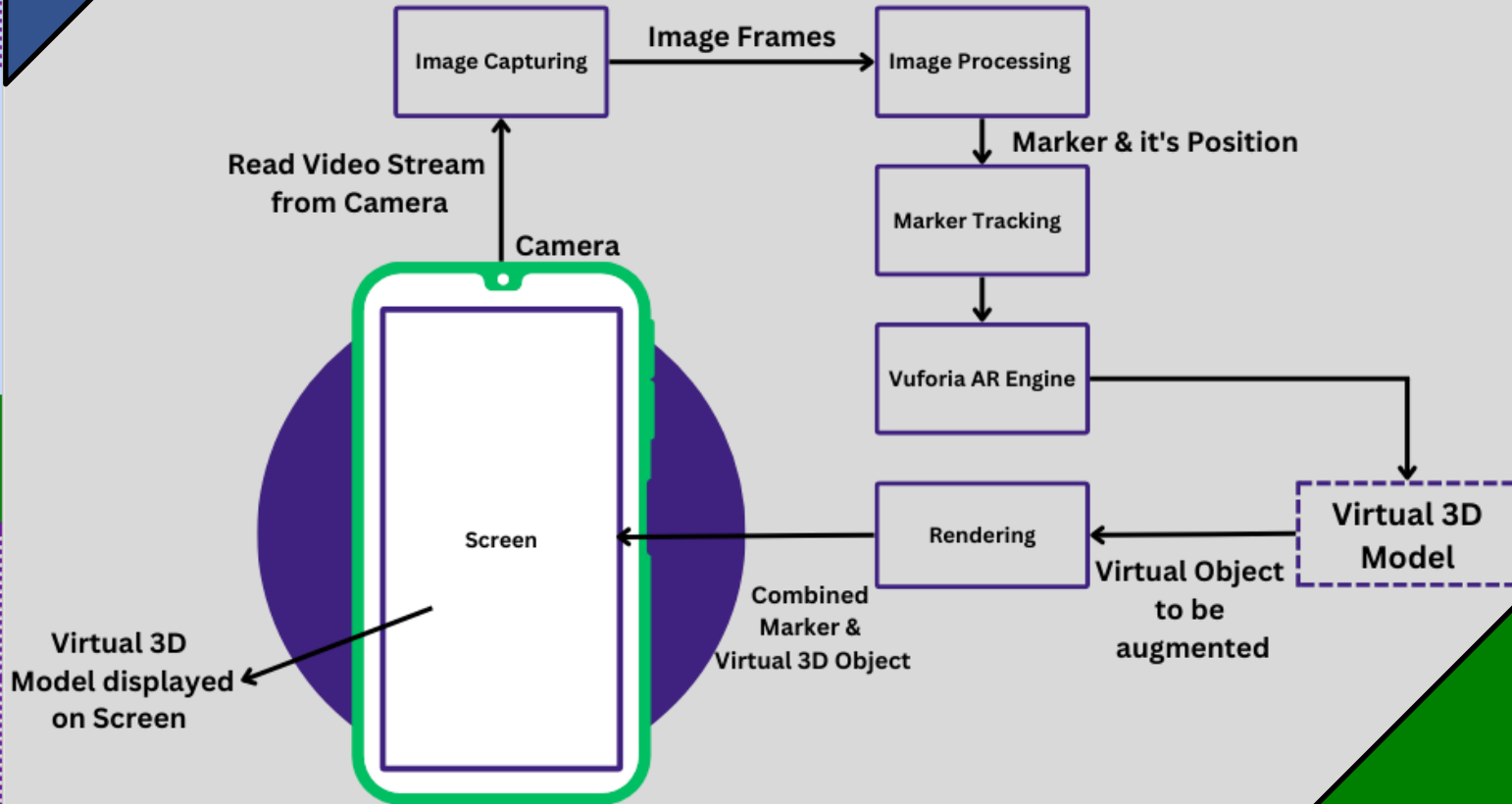
**CE - 119 : Computing Fundamentals (CF)**

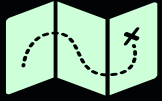
**Course Objectives:**

- This course covers the concepts and fundamentals of computing and programming. Topics includes history, components of computers, hardware, software, operating systems, database, networks, number systems and logic gates. Also it includes programming topics such as basic building blocks, loop, decision making statements.

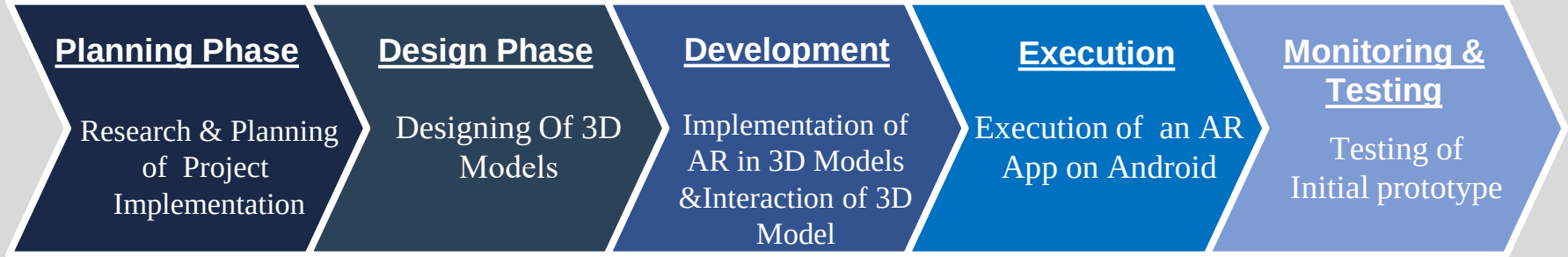


# System Functional Diagram





# Project Design Progress





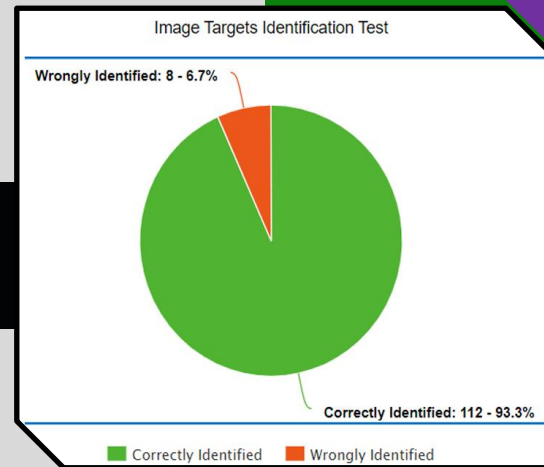
# Performance Metrics

| Rendering Aspect     | Performance Metric |
|----------------------|--------------------|
| Frame Rate           | 30 fps             |
| Model Loading Time   | 0.1-0.2 seconds    |
| Touch responsiveness | 0.15-0.2 seconds   |



# Experiments & Results

- We performed image targets identification tests
- 120 times on different devices
- We achieved accuracy of 93.3%





# Work Divisioning



Ramish Masood  
(2020-CE-086)

**Unity Developer/  
UI Designer**



Adil Aijaz  
(2020-CE-095)

**3D Models /  
Documentation**



Qasim  
(2020-CE-113)

**3D Models /  
Documentation**

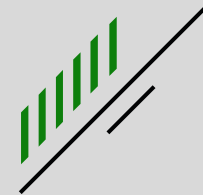


Zahir Khan  
(2020-CE-121)

**Unity Developer /  
Web Developer**



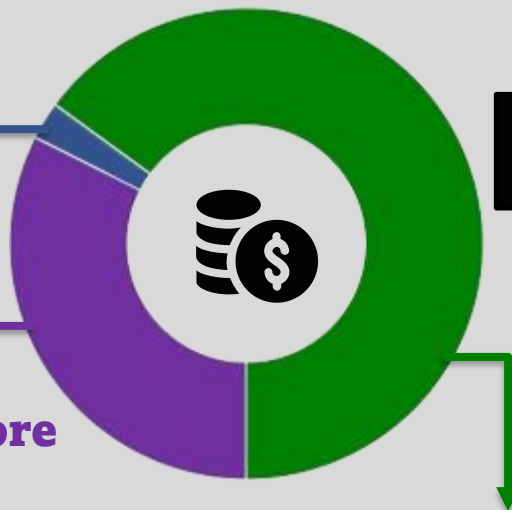
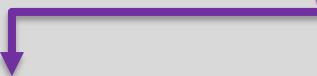
# Project Costing



**1.2% 3D Models**  
**2\$ (570 PKR)**



**14.1% Google Play Store**  
**25\$ (7132 PKR)**



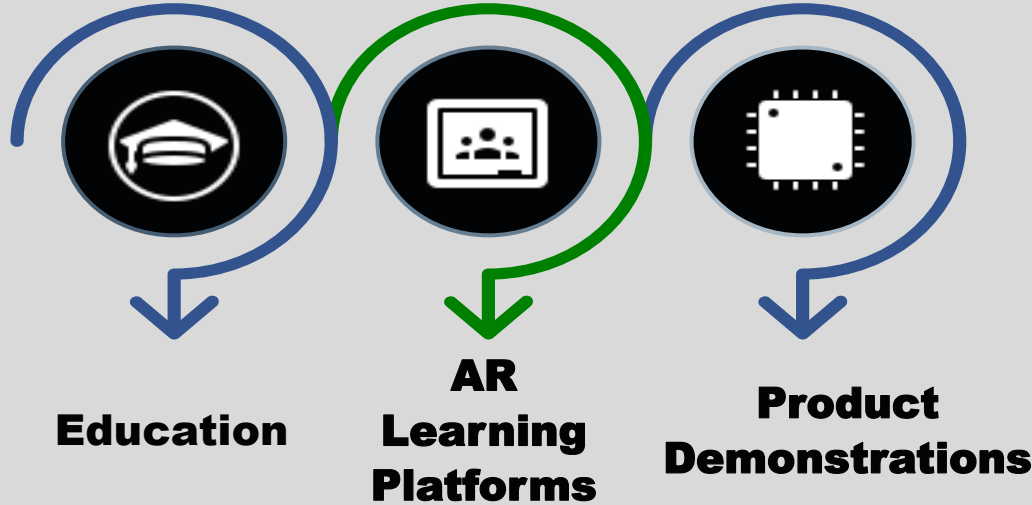
**Total Cost = \$77 (21507 PKR)**

**28.2% Miscellaneous**  
**50\$ (14265 PKR)**





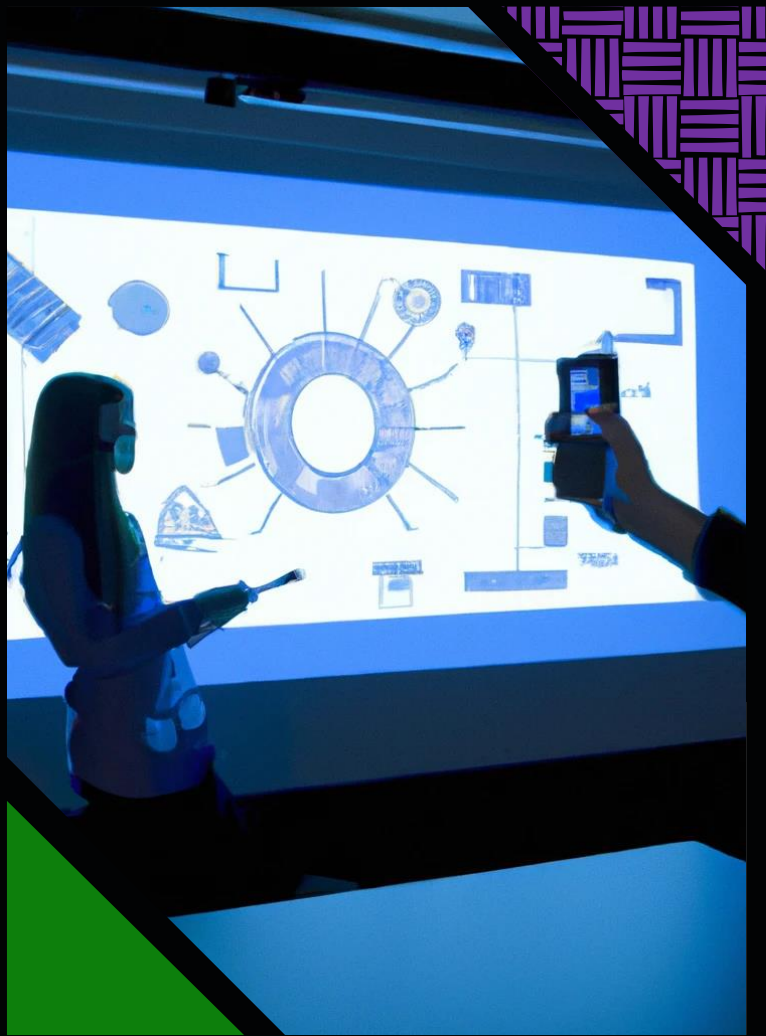
# Commercial Use





# Conclusion

Our "AR Based Computer Fundamentals Course App" presents a pioneering solution to the challenges of teaching computer hardware visualization. By leveraging the power of AR, our project offers an innovative and immersive learning experience, enabling students to interact with virtual computer components and overcome the limitations of traditional teaching methods. This fosters practical understanding and enhances the overall learning process.



*Thank you!*

